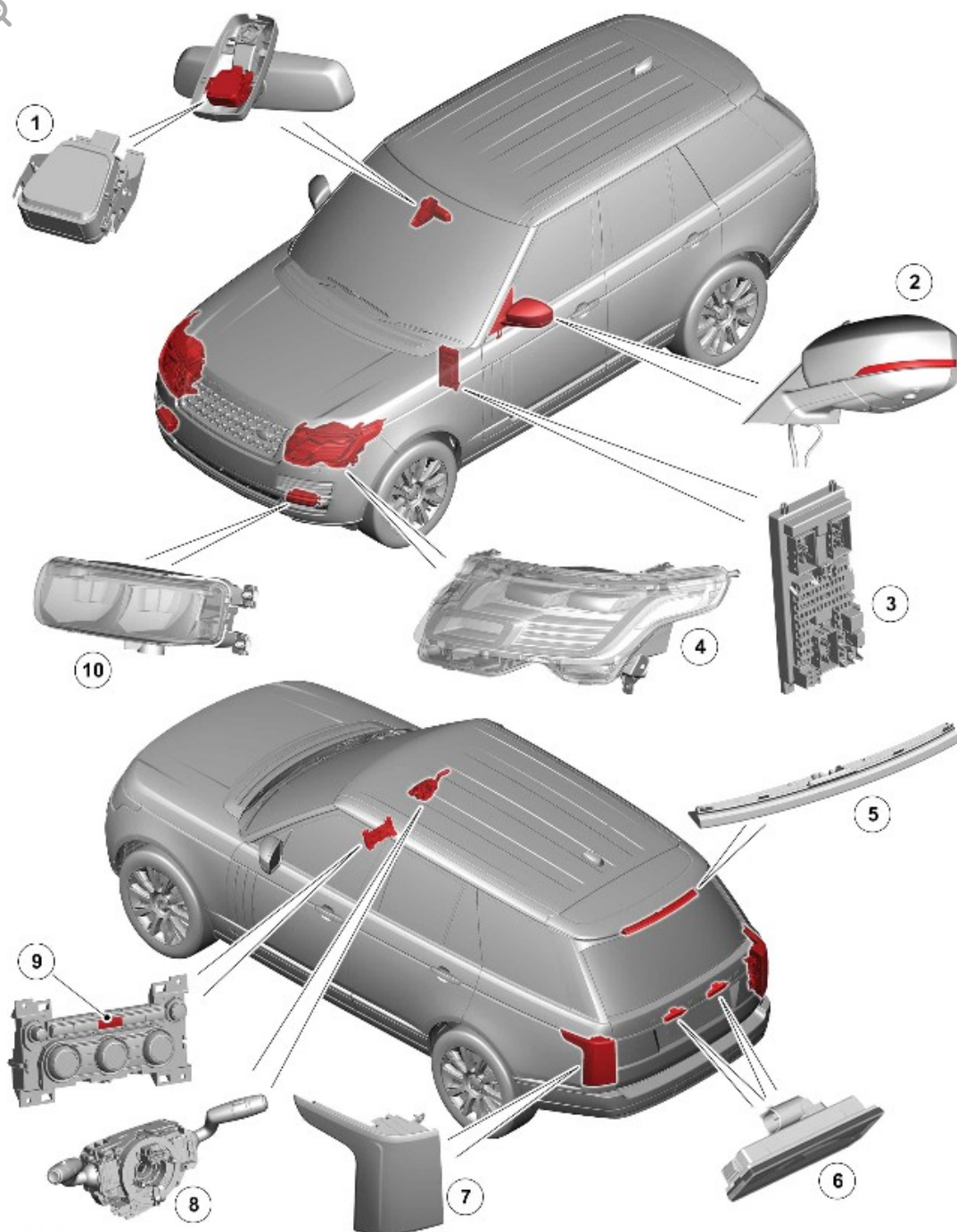


2016.0 RANGE ROVER (LG), 417-01

EXTERIOR LIGHTING

DESCRIPTION AND OPERATION

EXTERIOR LIGHTING COMPONENT
LOCATION



E204918

ITEM	DESCRIPTION
1	Interior mirror (Rain/light sensor and Auto High Beam (AHB) camera)
2	Side repeater lamp
3	Body Control Module/Gateway Module (BCM/GWM)
4	Headlamp
5	High Mounted Stop Lamp (HMSL)
6	License plate lamp

7	Tail lamp
8	Lighting control switch
9	Hazard warning lamp switch
10	Front fog lamp

OVERVIEW

The exterior lighting is controlled by the Body Control Module/Gateway Module (BCM/GWM). The BCM/GWM controls the following vehicle functions:

- Control and monitoring of exterior lamps including turn signal indicators and hazard warning functionality
- Communication and control and monitoring of trailer lighting via the trailer module
- Monitoring and evaluation of check control inputs from other system control modules and output of applicable messages in the Instrument Cluster (IC) message center.

The BCM/GWM is connected to the Medium Speed (MS) and High Speed (HS) Controller Area Network (CAN) bus. The BCM/GWM contains a microprocessor which performs the control, monitoring and evaluation functions.

A combined rain/light sensor is fitted which controls the automatic wiper operation and the automatic lighting function.

An Auto High Beam (AHB) system can also be fitted which automatically controls the high beam headlamps.

All exterior lamps are Light Emitting Diode (LED) apart from the Xenon headlamps.

The exterior lighting system comprises the following exterior lamps:

- Front and rear side lamps
- License plate lamps
- Side marker lamps (if fitted)
- Front and rear turn signal indicator lamps
- Turn signal indicators side repeater lamps
- Stop lamps and High Mounted Stop Lamp (HMSL)
- Reversing lamps
- Rear fog lamps
- Front fog lamps
- Static bending lamps (if fitted - Adaptive Front lighting System (AFS) headlamp except (North American Specification (NAS))
- Low and high beam headlamps
- Adaptive Front lighting System (AFS) (if fitted).
- Daytime Running Lamps (DRL)

DESCRIPTION

BODY CONTROL MODULE/GATEWAY MODULE

The Body Control Module/Gateway Module (BCM/GWM) is located behind the glove compartment and is connected to the vehicle wiring harness.

The BCM/GWM receives four permanent battery power supplies via the Battery Junction Box (BJB)

The lighting circuits are not all protected by conventional fuses as some are protected by Metal Oxide Semiconductor Field Effect Transistors (MOSFETs). The control circuitry within the BCM/GWM and the BCM/GWM for each individual circuit can detect and isolate a problem circuit.

Failure of a lamp is not notified to the driver. If a turn signal indicator fails it will flash at double speed.

INPUT SIGNALS FOR LAMP CONTROL

The Body Control Module/Gateway Module (BCM/GWM) receives inputs from the following switches:

- Left hand steering column multifunction switch for turn signal indicators, main beam, front and rear fog lamps, side lamps, headlamps, auto headlamps, high beam/headlamp flash and Auto High Beam system
- Brake pedal switch
- Momentary push switch for hazard warning.

The BCM/GWM also receives ignition status via hard wired connections from the stop/start switch.

A reverse gear engaged signal is also received on the High Speed (HS) Controller Area Network (CAN) powertrain bus from the Transmission Control Module (TCM) to enable the BCM/GWM to activate the reverse lamps.

The BCM/GWM can receive a hazard warning indicator activation message from the Restraints Control Module (RCM), via the HS CAN powertrain bus, in the event of a crash. The BCM/GWM can also activate the hazard warning indicators to signify vehicle locking to the driver.

On vehicles with Auto High Beam (AHB), the Auto High Beam Control Module (AHBCM) outputs signals on the HS CAN powertrain bus to the BCM/GWM to control the high beam headlamps.

CIRCUIT PROTECTION

Operation of the lamps is performed using Metal Oxide Semiconductor Field Effect Transistors (MOSFETs). The MOSFETs can detect overload, load interruption with the lamps switched on and short circuit to positive with the lamps switched off.

The MOSFETs are protected against short circuits, removing the requirement for the lamps circuits to be protected by fuses. The MOSFETs respond to heat generated by increased current flow caused by a short circuit. Normally this would cause the fuse to blow. The MOSFETs react to the heat increase and cut the supply to the affected circuit. Once the fault has been rectified or the MOSFET has cooled, the MOSFET will automatically reset and operate the circuit normally.

If an overload occurs, the current flow is dependant on the temperature of the related MOSFET and can be up to 20 times the rated current of the lamp. The MOSFET heats up and deactivates the load applied to the circuit. When the MOSFET cools the circuit is once again reactivated. This thermal cycling occurs continuously in the event of an overload occurring.

The front fog lamps are controlled by relays and the circuit is protected by conventional fuses.

XENON BULB CONTROL MODULE

If a xenon draws high current the Body Control Module/Gateway Module (BCM/GWM) will detect a short circuit and remove the supply to the Xenon Bulb Control Module (XBCM). If the XBCM unit is disconnected then the BCM/GWM will continue to power the XBCM as there is no open load detection enabled for the Xenon output.

ALARM INDICATIONS

The Body Control Module/Gateway Module (BCM/GWM) can also display alarm visual indications for alarm arm, disarm and triggered conditions.

If the hazard warning lamps are active when a lock or unlock request is made, the hazard warning cycle is interrupted to allow the visual indication of the requested lock cycle. When visual indication is completed, the hazard warning operation will continue.

If the vehicle is involved in crash of a severity for the Restraints Control Module (RCM) to initiate deployment of the airbags, the control module outputs a hazard warning lamps on request on the Medium Speed (MS)

Controller Area Network (CAN) bus to the BCM/GWM. The hazard warning lamps will be activated and will continue until the RCM outputs a message to deactivate the hazard warning lamps.

REDUNDANT DATA STORAGE

The Body Control Module/Gateway Module (BCM/GWM) stores data relating to the Vehicle Identification Number (VIN), total mileage and service interval indicator. This data is received by the BCM/GWM from the Instrument Cluster (IC) and used as a back-up in the event of IC replacement.

If the BCM/GWM is to be replaced, an approved diagnostic system must be connected to the vehicle and the BCM/GWM replacement procedure followed to ensure that the stored data is transferred to the new unit.

CRASH SIGNAL ACTIVATION

In the event of an accident of a severity to activate and deploy the airbags, the Restraints Control Module (RCM) requests various electrical operations to assist with the crash situation. The RCM requests via the bus systems to the Body Control Module/Gateway Module (BCM/GWM) to activate the hazard warning lamps.

SECURITY SYSTEM ACTIVATION

In the event of the security system being triggered, the Body Control Module/Gateway Module (BCM/GWM) requests activation of the hazard warning lamps.

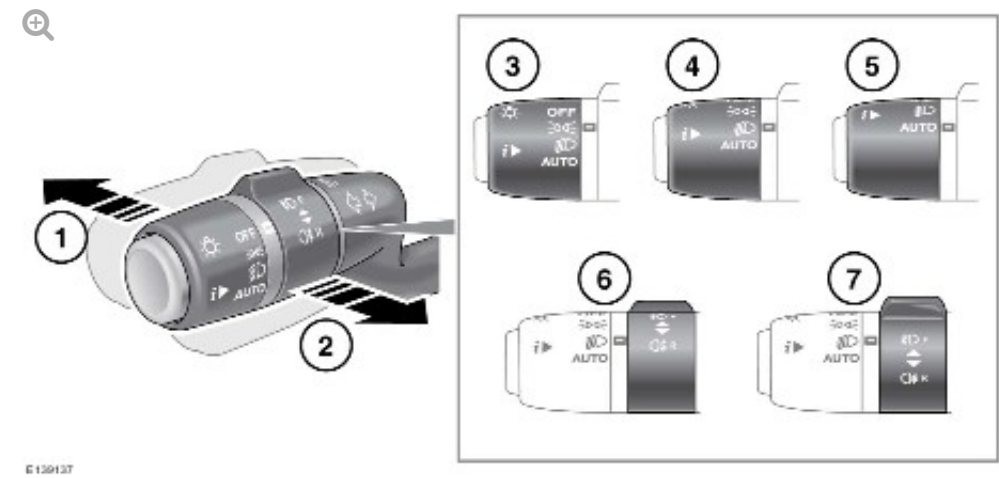
INSTRUMENT PANEL AND SWITCH ILLUMINATION DIMMING

The Body Control Module/Gateway Module (BCM/GWM) controls the instrument cluster backlighting illumination and also illumination of all instrument panel switches.

The BCM/GWM supplies a power output to all switch illumination at a voltage determined by the position of the manual dimmer rheostat. The switch illumination is activated when the lighting control switch is in the side lamp or headlamp position. The switch illumination can also be controlled

via the dimmer when the door is opened and the vehicle is unlocked.

LIGHTING CONTROL SWITCH



ITEM	DESCRIPTION
1	High Beam select
2	High beam cancel/flash
3	Side lamp on
4	Headlamp on
5	Auto headlamp position
6	Rear fog lamp on/off
7	Front fog lamp on/off

All lighting (apart from the hazard warning lamps) control is via the left hand steering column multifunction switch. The switch comprises two rotary switches for controlling side lamps and headlamps and fog lamps and a push/pull function for high/low beam selection.

The lighting control switch is hardwired to the steering wheel module which in turn is connected to the Body Control Module/Gateway Module (BCM/GWM) via a Local Interconnect Network (LIN) bus.

The left hand steering column multifunction switch is located on the left hand side of the steering column and controls the following functions:

- Headlamp low/high beam
- Headlamp high beam flash
- Left/right turn signal indicators
- Trip computer functions. For additional information refer to 413-08 Information and Message Center.

The high beam on and flash functions are connected on a single wire to the BCM/GWM. The switch assembly contains resistors and the ground for each selection is passed through a different resistor for each selection. The same also applies to the turn signal indicators. The BCM/GWM senses the resistance on the ground wire and determines which function has been selected.

The turn signal indicators incorporate a lane change function. To activate the lane change function, hold the lever partially up or down against spring pressure to indicate a lane change. The directional indicator will flash three times. The lane change function can be enabled or disabled using an approved diagnostic system.

HEADLAMP ASSEMBLY

GENERAL

Two types of headlamp are available; Bi-xenon™ or Adaptive Front lighting System (AFS).

Headlamp removal is not required for replacement of the xenon bulbs. The rear of the headlamp unit has a removable access cover which allow access to the bulb.

A large cover, which is rotated counter-clockwise to remove, allows access to the low/high beam xenon bulb.

The turn signal indicators, side lamps, static bending lamp (where fitted), daylight running lamps and the side marker lamp (where fitted) are Light Emitting Diode (LEDs). None of the LEDs are serviceable items.

In all markets except North American Specification (NAS), the headlamps have two adjustment screws which allow for the manual setting of the vertical and horizontal beam alignment. A 6mm Allen key is used to rotate the adjusters to achieve the required setting. The inboard adjuster controls the vertical aiming and the outboard adjuster controls the horizontal aiming.

On NAS vehicles the headlamp is regarded as 'Visual Optically Left' (VOL) aiming. There is no horizontal adjustment. Refer to the Service Repair Procedures manual for headlamp alignment data and procedures.

Each headlamp has an integral sixteen pin connector which provides inputs and outputs for the various functions of the headlamp assembly. The usage of the pins differs between model variants. Refer to the Electrical Library and circuit diagrams for pin details.

Two breathers are located on the rear of the headlamp housing. The Goretex breathers allow air flow in and out of the headlamp but prevent the ingress of moisture from rain, road spray or washing. If condensation occurs within the headlamp, check that the breathers are not blocked with mud etc before further action is taken.

HEADLAMP OPERATION

The lamps contained within the headlamp assembly have differing functionality depending on the function selected.

The low beam headlamps are switched on when the ignition is in power mode 6 (ignition on) and:

- the lighting control switch is the headlamp position or
- the lighting control switch is in the 'AUTO' position and a 'lights on' signal is received by the Body Control Module/Gateway Module (BCM/GWM) from the rain/light sensor.

The low beam headlamps can also be operated by the headlamp delay feature.

The high beam headlamps are switched on when the ignition is in power mode 6 (ignition on) and:

- the lighting control switch is in the headlamp position or the headlamps are activated by the AUTO feature and the left hand steering column multifunction switch is pushed forward, away from the driver or
- the headlamp flash function is operated by pulling the left hand steering column multifunction switch towards the driver or
- the Auto High Beam (AHB) system is active.

COMMON HEADLAMP FEATURES

Turn Signal Indicators

The turn signal indicator Light Emitting Diodes (LED) are in the outer part of the headlamp assembly. The LEDs are not serviceable components. The LEDs are controlled by an LED control module located at the rear of the headlamp assembly.

Side Lamps

On the low line headlamp design the side lamps and daytime lamps are located in the inner lamp of the headlamp, on High line they are located in a pattern around the outer edge of the inner lamp and along the lower edge of the lamp unit. The Light Emitting Diode (LED)s are not serviceable components. The LEDs are controlled an LED control module located at the rear of the headlamp assembly.

Side Marker Lamps (NAS only)

The side marker lamp Light Emitting Diode (LED) high is located in a molded receptacle in the outer edge of the headlamp. The side marker lamp illuminates a small rectangular section between the orange side reflectors on the outside of the headlamp. The LED is not a serviceable

component.

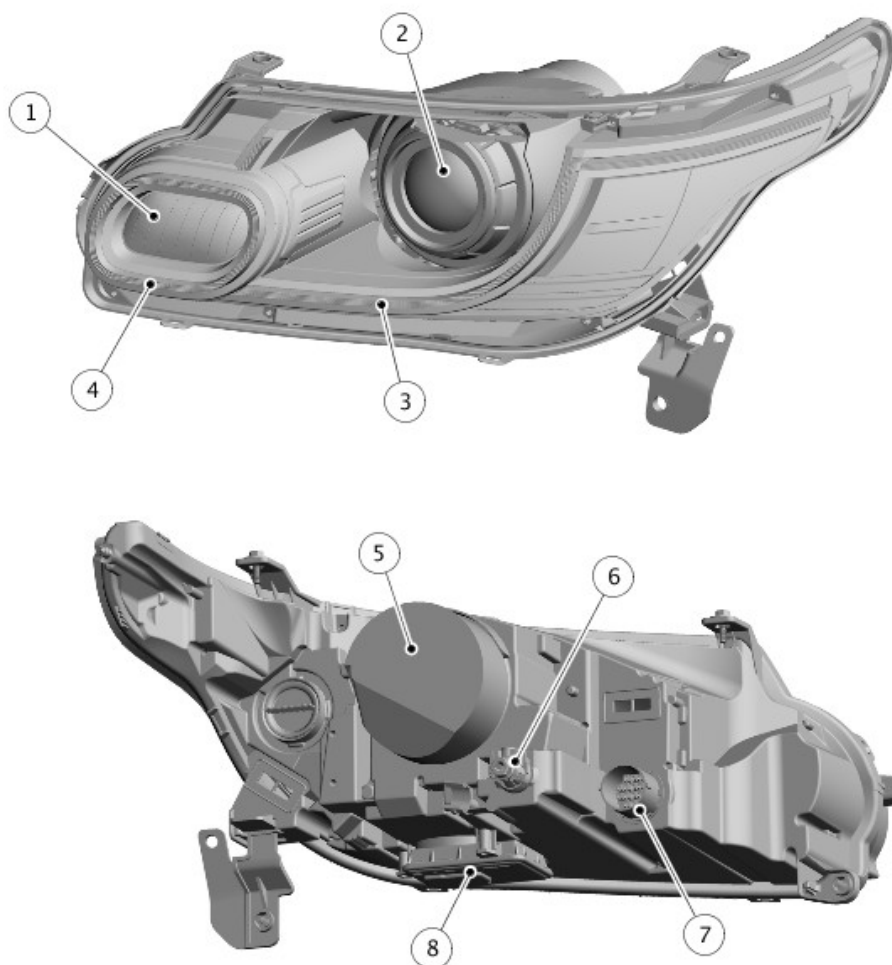
Headlamp Hand of drive

Previous models required the use of a tourist lever to set the headlamps for driving on a particular side of the road. The Range Rover has a headlamp beam pattern that negates the need for a mechanical lever. Instead the headlamps can be configured for the side of road on which the vehicle is to be driven on via the Vehicle set-up in the message centre. When in vehicle set-up select high beam assist, then select the side of road that the vehicle is to be driven on.

Auto High Beam

Instead the headlamps can be configured for the side of road on which the vehicle is to be driven on via the Vehicle set-up in the message center. When in vehicle set-up select Auto High Beam (AHB), then select the side of road that the vehicle is to be driven on.

XENON HEADLAMPS



E156123

ITEM	DESCRIPTION
1	Daylight running lamp and side lamp (Low line) cornering lamp (High line)
2	Bi-xenon™ projector module (dipped beam and high beam)
3	Turn signal indicator Light Emitting Diode (LED)
4	Daylight running and side lamps (High line headlamps)
5	Cover - bi-xenon™ headlamp bulb
6	Horizontal beam adjuster
7	Electrical connector
8	Light Emitting Diode Driver Module (LEDDM)

Safety Precautions

WARNING:

The Xenon system generates up to 28000 volts and contact with this voltage could lead to fatality. Make sure that the headlamps are switched off before working on the system.

The following safety precautions must be followed when working on the xenon headlamp system:

- **DO NOT** attempt any procedures on the xenon headlamps when the lights are switched on
- Handling of the D3S xenon bulb must be performed using suitable protective equipment, e.g. gloves and goggles. The glass part of the bulb must not be touched
- Xenon bulbs must be disposed of as hazardous waste
- Only operate the lamp in a mounted condition in the reflector.

The headlamps use a complex surface reflector for the halogen high beam 'fill-in' lamp. This type of lamp has the reflector divided into separate parabolic segments, with each segment having a different focal length. A halogen H7LL 55W bulb is retained in an extended holder. The holder is secured in the rear of the high beam 'fill-in' lamp lens by rotating clockwise.

The xenon headlamp is known as 'Bi-Xenon™' (the Bi-Xenon™ trademark is the property of Hella KGaA Hueck & Co., Germany) because it operates as both a low and high beam unit. The xenon bulb is located in a projector module which comprises an ellipsoidal lens with a solenoid controlled shutter to change the beam output from low to high.

NOTE:

If the lighting control switch is in the 0 (off) position, the xenon lamps will operate when the high beam 'flash' function is operated, the lamp will be lit but the shutter will only operate in high beam. If the lighting switch is in the headlamp position or the 'AUTO' position with the low beam headlamp active, the xenon low beam will remain on when the high beam 'flash' function is operated.

The xenon headlamp system is controlled by the Body Control Module/Gateway Module (BCM/GWM) using a Xenon Bulb Control Module (XBCM) and an igniter for each headlamp. The XBCM and the igniters provide the regulated power supply required to illuminate the xenon bulbs through their start-up phases of operation.

The xenon bulb is located in the rear of the projector module. The D3S xenon bulb incorporates an integral igniter unit and both components must be replaced if the bulb fails.

The xenon headlamp is a self contained unit located within the headlamp assembly. The unit comprises a reflector, an adaptor ring, the lens, a shutter controller and the xenon bulb, which as an assembly is known as the projector module.

The reflector is curved and provides the mounting for the xenon bulb. The bulb locates in a keyway to ensure correct alignment in the reflector. The bulb is an integral part of the igniter and is electrically connected by a connector located in the igniter unit.

The shutter controller is a solenoid which operates the shutter mechanism via a lever. The shutter is used to change the beam projection from low beam to high beam and visa versa.

The xenon bulbs illuminate when an arc of electrical current is established between two electrodes within the bulb. The xenon gas sealed in the bulb

reacts to the electrical excitation and the heat generated by the current flow to produce the characteristic blue/white light.

To operate at full efficiency, the xenon bulb goes through three stages of operation before full output for continuous operation is achieved. The three phases are; start-up phase, warm-up phase and continuous phase.

In the start-up phase, the bulb requires an initial high voltage starting pulse of up to 30000 volts to establish the arc. This is produced by the igniter. The warm-up phase begins once the arc is established. The XBCM regulates the supply to the bulb to 2.6A which gives a lamp output of 75W. During this phase, the xenon gas begins to illuminate brightly and the environment within the bulb stabilizes ensuring a continual current flow between the electrodes. When the warm-up phase is complete, the XBCM changes to continuous phase. The supply voltage to the bulb is reduced and the operating power required for continual operation is reduced to 35W. The process from start-up to continuous phase is completed in a very short time.

The xenon system is controlled by the Body Control Module/Gateway Module (BCM/GWM), the two Xenon Bulb Control Module (XBCM) and the two igniters. The XBCM (one per headlamp) receive an operating voltage from the BCM/GWM when the headlamps are switched on. The modules regulate the power supply required through the phases of start-up.

The igniters (one per headlamp) generate the initial high voltage required to establish the arc. The igniters have integral coils which generate high voltage pulses required for start-up. Once the xenon bulbs are operating, the igniters provide a closed circuit for regulated power supply from the control modules.

ADAPTIVE FRONT LIGHTING SYSTEM HEADLAMPS

The Adaptive Front Lighting System (AFS) headlamp is similar in its construction to the xenon headlamp described previously. The projector module is constructed and operates as described for the xenon headlamp with the addition of the AFS system which allows the projector module to be moved vertically and horizontally by stepper motors. The following

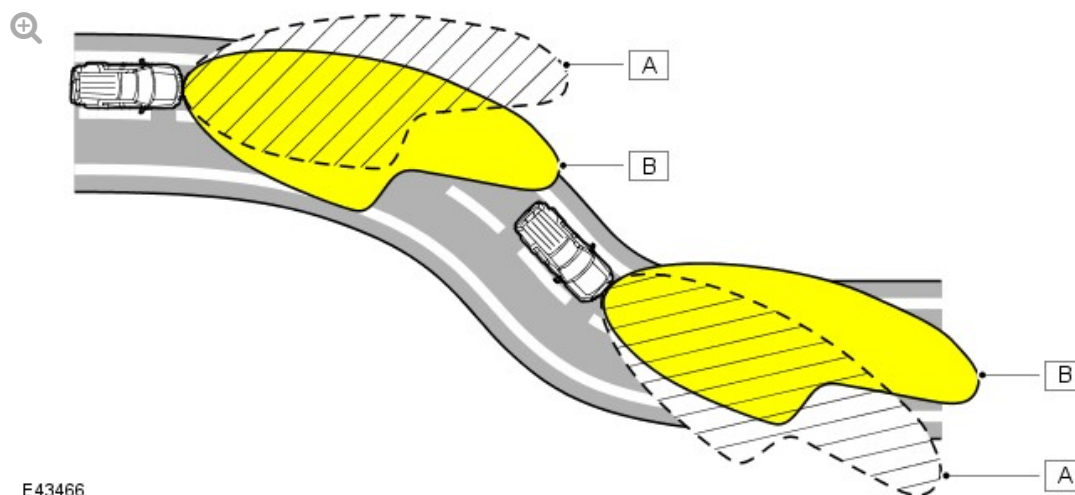
description covers the additional differences to the xenon headlamp with AFS.

The AFS is a system to improve driver visibility under differing driving conditions. AFS provides a larger visible area which is illuminated when cornering by adjusting the position of the beam distribution on the road. Horizontal adjustment is made automatically to the most suitable orientation for the driving conditions using steering angle and information from other vehicle sensors.

AFS includes the automatic headlamp levelling system described in the 'Headlamp Levelling' section of this document. The bi-xenon™ module within the headlamp is controlled by actuator motors which rotate the projector module on its vertical and horizontal axes to adjust the beam output to suit the cornering conditions and vehicle inclination.

The AFS is controlled by an AFS control module which is integrated into the Body Control Module/Gateway Module (BCM/GWM) The module is connected to and controls an AFS power module located inside the headlamp housing. Signals from the AFS control module are processed by the AFS power module which powers stepper motors to adjust the vertical and horizontal alignment of the projector module. The AFS power module also controls and regulates the operation of the static bending lamp (if fitted) which is requested by the AFS control module but controlled by the Body Control Module/Gateway Module (BCM/GWM).

AFS Concept



ITEM	DESCRIPTION
A	Conventional headlamp beam distribution
B	Adaptive Front Lighting System (AFS) headlamp beam distribution

The Adaptive Front Lighting System (AFS) xenon headlamp construction is similar to the non-AFS xenon headlamp assembly. The AFS headlamp has a Xenon Bulb Control Module (XBCM) located on the underside of the lamp assembly. An additional AFS power module is located inside the headlamp housing. The AFS power modules supply the correct voltage to the stepper motors which control the positioning and movement of the AFS projector module.

The AFS assembly contains an additional carrier frame which provides the location for the AFS actuators. The remaining lamps are as described previously for the xenon headlamp. The AFS headlamp also incorporates a static bending/cornering lamp (except on North American Specification (NAS) market vehicles).

The carrier frame is attached to the AFS vertical actuator. The projector module has a central pivot point which allows the module to move horizontally in response to operation of the AFS horizontal actuator.

The AFS actuators are bi-polar (2 phase) dc stepper motors which are driven by a power output from the AFS power module. Each stepper motor

receives its position information from the AFS control module via the applicable AFS power module. When the actuators are powered to their requested positions, a holding current is applied to maintain the actuator position.

The actuators do not supply a positional feedback signal to the AFS control module. Each stepper motor requires referencing each time the AFS system becomes active. When the AFS system is active, each actuator is driven a small amount. The low beam is driven to just below horizontal after headlight levelling is complete. The horizontal actuator is driven a small amount inboard and then a small amount outboard.

HEADLAMP LEVELLING CONTROL MODULE

The Headlamp Levelling Control Module (HLCM) is also integrated into the Body Control Module/Gateway Module (BCM/GWM).

The HLCM is a dual functionality unit which also incorporates software to control the automatic headlamp levelling. The headlamp levelling control module is connected to the High Speed (HS) Controller Area Network (CAN) powertrain bus and receives inputs from other vehicle systems on the status of the following parameters:

- Steering angle
- Vehicle speed
- Headlamp status
- Engine running
- Reverse gear selected
- Automatic lighting on.

The Adaptive Front Lighting System (AFS) will only operate when the HLCM receives an engine running signal on the CAN bus. When the engine running signal is received the HLCM performs an initialization routine.

The AFS will also function when the lighting control switch is in the AUTO

position and the HLCM receives a lights on signal from the rain/light sensor and an engine running signal.

The HLCM then monitors the inputs from the other vehicle systems to control the AFS functionality according to cornering (steering) angles and vehicle speed.

The HLCM is connected to each AFS power module on a private Local Interconnect Network (LIN) bus. The power modules read operating values supplied from the HLCM and control the output drivers for the stepper motor actuators inside the headlamp assembly.

AFS OPERATION

The Adaptive Front Lighting System (AFS) controls the swivelling angle of each projector module using speed and steering angle signals. The angles of each projector module differ to give the correct spread of light, e.g. when turning left, the left projector module will have a greater swivelling angle than the right hand projector module.

INITIALIZATION PROCEDURE

When the Adaptive Front Lighting System (AFS) control module receives an ignition on signal, the control module performs the initialization procedure which ensures that the headlamps are correctly aligned on both their vertical and horizontal axes.

The AFS swivel initialization starts less than one second after the headlamp levelling initialization is activated to ensure that the headlamps are at or below the 0 degree position in the vertical axis, thus preventing glare to oncoming vehicles. The AFS swivel initialization is completed in less than 2.5 seconds. The low beam is driven to just below horizontal after headlight levelling is complete. The left and right AFS actuator motors are powered from the 0 degree position to a small movement to the inboard position, then another small movement to the outboard position and then back to the 0 degree position.

FAILURE MODE

In the event of a failure of the , a warning indicator in the Instrument Cluster (IC) is illuminated to warn the driver. The Adaptive Front Lighting System (AFS) warning indicator illuminates when the ignition is in power mode 6 (ignition on) and will flash continuously until the fault is rectified. The AFS warning indicator will also be illuminated if a failure of the steering angle sensor or the vehicle speed signal is detected.

Illumination of the AFS warning indicator does not necessarily mean that there is a fault with the AFS. The fault may be caused by a failure of another system preventing the AFS system operating correctly.

The AFS control module performs a diagnostic routine every time AFS is requested. If any fault is found, the AFS control module will suspend the operation of the AFS function.

If the AFS levelling system has failed with the xenon projector module in a position other than the correct straight ahead position, the AFS control module will attempt to drive the projector module to a position a small amount lower than the standard position. If the swivel function has failed, the AFS control module will lower the projector module using the levelling actuator motors to a position much lower than standard to prevent excess glare to oncoming vehicles.

The AFS control module software can detect an internal failure of the control module control circuits. The control module will power the projector modules to the zero position and prevent further operation.

Faults can be investigated by interrogating the AFS control module using the recommended diagnostic tool to check for fault codes.

STATIC BENDING/CORNERING LAMPS

NOTE:

The static bending/cornering lamps are not fitted to North American Specification (NAS) vehicles with Adaptive Front Lighting System (AFS) headlamps.

The static bending/cornering lamps, which are a standard feature on Adaptive Front Lighting System (AFS) headlamps, are designed to illuminate the direction of travel when cornering at low speeds. The static bending/cornering lamp functionality, which is controlled by the Body Control Module/Gateway Module (BCM/GWM), is unique to vehicles with AFS headlamps and operates using inputs from the steering angle sensor.

The static bending/cornering lamp Light Emitting Diode (LED)s are incorporated into the headlamp assembly. The design of the lens projects a spread of light from the vehicle at approximately 45 degrees to the vehicle axis.

The static bending/cornering lamp uses 2 high power LEDs located in the headlamp housing. The LEDs are not serviceable components.

Cornering Lamp Functionality

The cornering lamps are designed to illuminate the direction of travel when cornering at low speeds. The design of the lens projects a spread of light from the vehicle at approximately 45 degrees to the vehicle axis.

The cornering lamps are controlled by the left steering column multifunction switch with the lighting control switch in the headlamp position and the ignition switch in power mode 6 (ignition on). The cornering lamps are not supplied power when the vehicle is in power mode 6 to ensure that they do not function with the headlamp delay feature. The cornering lamps are deactivated if the vehicle speed exceeds 25 mph (40 km/h) at which point the static bending lamp functionality is activated. Cornering lamps only illuminate when:

- In Drive
- Vehicle speed is slower than 40kph
- One of the direction indicators is on
- Headlamps are dipped

Only one cornering lamp will illuminate at any one time. If the left turn signal indicators are selected on, the left hand cornering lamp will be illuminated and visa versa, providing the vehicle speed and lighting control switch positions are correct.

Static Bending Lamp Functionality

The static bending lamps operate with a steering angle sensor Controller Area Network (CAN) signal and vehicle speed signal which is received by the Adaptive Front Lighting System (AFS) control module and the Body Control Module/Gateway Module (BCM/GWM). The AFS control module sends a static bending lamp on request to the BCM/GWM which activates the static bending lamp Light Emitting Diode (LED)s.

When the operation parameters of the lamp are reached, the BCM/GWM illuminates the static bending lamp LEDs on using a full power Pulse Width Modulation (PWM) voltage. When the lamp is switched off, the BCM/GWM fades the LEDs off by decreasing the PWM voltage in a linear manner.

AUTOLAMPS

The Autolamp function is a driver assistance system. The driver can override the system operation by selection of side lamps or headlamps on if the ambient light conditions require front and rear lighting to be active.

The Autolamp system uses a rain/light sensor which is connected via the Local Interconnect Network (LIN) bus to the Body Control Module/Gateway Module (BCM/GWM) to control the headlamp functionality. The light sensor is incorporated in the rain/light sensor located on the inside of the windshield, below the rear view mirror. The wiper system also uses the

rain/light sensor for automatic wiper operation. Refer to the Wipers and Washers section for details of the rain/light sensor and automatic wiper operation. For additional information refer to Wipers and Washers 501-16.

The rain/light sensor measures the ambient light around the vehicle in a vertical direction and also the angular light level from the front of the vehicle. The rain/light sensor uses vehicle speed signals, wiper switch position and the park position of the front wipers to control the system.

The rain/light sensor can detect if the vehicle has entered a tunnel or similar environment and will activate the headlamps on entry to the tunnel when the ambient and forward light levels have fallen quickly. When the tunnel is exited, the rain/light sensor detects the sudden increase in light levels and requests the lights to be switched off.

Certain light and weather conditions are not detected accurately by the rain/light sensor. The driver should override the automatic headlamps function if in any doubt about weather conditions and the requirement for exterior lights to be active.

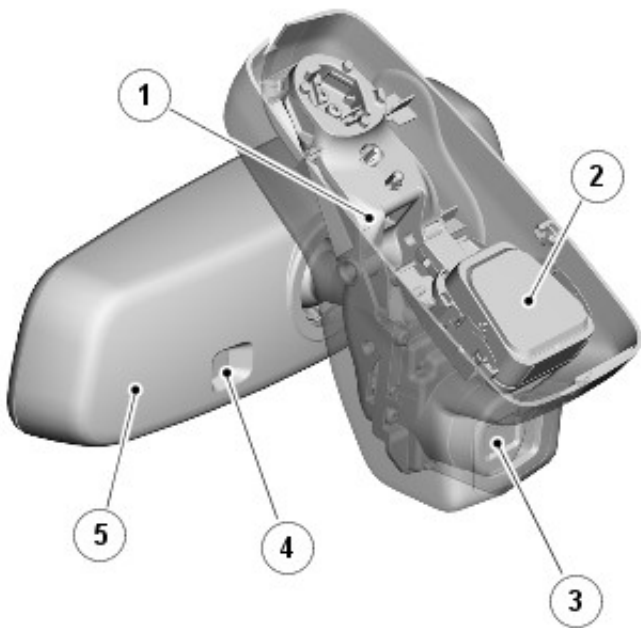
The automatic headlamp operation uses ambient light levels which are monitored by a photodiode incorporated in the rain/light sensor. The rain/light sensor sends a lights on/off request to the BCM/GWM via the LIN bus, which responds by switching on the low beam headlamps, front side lamps and rear tail lamps. The Autolamps are activated under the following conditions:

- Twilight
- Darkness
- Rain
- Snow
- Tunnels
- Underground or multi-level car parks.

Operation of the Autolamps requires the ignition to be in power mode 6 (ignition on), the lighting control switch to be in the 'AUTO' position and a lights on request signal from the light sensor.

If the Autolamps function has been selected and the ambient light falls below a pre-defined level then the front and rear fog lamps can be manually activated. If the ambient light rises above that level then the fog lamps will be deactivated along with the rest of the lamps. If the ambient light then falls below this level again the lamps will be activated, but the fog lamps which were previously selected will not.

AUTO HIGH BEAM ASSIST



E148927

ITEM	DESCRIPTION
1	Rear view mirror calibration bracket
2	Rain/light sensor (Autolamps)

3	Image sensor
4	Front light sensor
5	Auto High Beam Control Module (AHBCM)

Auto High Beam (AHB) is a driving aid which automatically controls the high beam function. If required, the system can be overridden by the driver.

CAUTION:

The AHB is a driving aid only. Should the road conditions require, it is the drivers responsibility to consider other road users and operate the high beam headlights in a safe manner. In certain circumstances the driver will be required to intervene.

Auto High Beam Warning Indicator



E117699

ITEM	DESCRIPTION
1	Warning indicator (green)

The Auto High Beam (AHB) system is controlled by an Auto High Beam Control Module (AHBCM) which is located in the interior rear view mirror body and by the Body Control Module/Gateway Module (BCM/GWM). The module and the BCM/GWM are connected via the Medium Speed (MS)

Controller Area Network (CAN).

The AHBCM receives a power supply from the BCM/GWM when the ignition is in power mode 6 (ignition on). The rear view mirror also includes a low resolution camera (image) sensor which detects headlamps and tail lamps of preceding vehicles. The sensor is connected to the control module which evaluates the image data, checking for light intensity and location.

If conditions are correct, the control module will activate the high beam assist by sending a high or low beam request message to the BCM/GWM via the High Speed (HS) Controller Area Network (CAN) powertrain bus. The BCM/GWM then controls the shutter in the Xenon projector module.

AUTO HIGH BEAM OPERATION

The Auto High Beam (AHB) operates as part of the automatic headlight system. When driving at night with the lighting control switch in the automatic position and the left steering column multifunction switch in the central position, with sufficient darkness (approximately 1 lux or less) and a suitable road speed, the AHB will automatically operate the high beam lighting when necessary. A warning symbol in the instrument cluster confirms to the driver when the AHB is selected and enabled.

NOTES:

- The function of the normal 'blue' high beam indicator remains unchanged and it always reflects the actual status of the high beam lamps
- The exterior lighting 'on' threshold for the auto headlamps system is approximately 100 lux which is measured by the rain/light sensor. At light levels below this value the low beam headlamps and exterior lights will be switched on. The high beam assist will not function until the light level has reached approximately 1 lux. At light levels above 1 lux high beam is not required and therefore is not activated.

Activation (system ready)

The Auto High Beam (AHB) will only activate and illuminate the warning indicator to show system is ready or 'primed' for high beam control, when the following conditions are met:

- AHB has been first 'enabled' via the instrument cluster menu
- Lighting control switch is in the 'Auto' position
- Left steering column multifunction switch in the central position
- The ambient light level is below 100 lux – refer to 'Light Levels' section that follows
- The system has not been overridden or cancelled – refer to 'Override' section that follows
- The camera (image) sensor view is not blocked.

High Beam Control

When activated, Auto High Beam (AHB) will switch the headlamps to high beam when all the following conditions occur:

- No relevant oncoming traffic
- No relevant preceding traffic
- In non-urban environment, i.e. with no street lighting
- Ambient light level is below 1 lux – refer to 'Light Levels' section that follows
- Road speed is suitable – refer to 'Road Speed' section that follows.

Low Beam Control

When activated, high beam assist will switch the headlamps to low beam when any of the following conditions occur:

- Relevant oncoming traffic is present

- Relevant preceding traffic is present
- In urban environment, i.e. with street lighting
- Ambient light level is above 1 lux – refer to 'Light Levels' section that follows
- Road speed is not suitable – refer to 'Road Speed' section that follows
- Unrecognisable reflective inputs from road signs or markings – refer to 'System Limitations' section that follows.

Light Levels

The exterior lighting 'on' threshold for the normal 'auto headlamps' feature is approximately 100 lux and is measured by the windscreen mounted 'rain/light' sensor. When the light level falls to this value the low beam headlamps and exterior lights will be switched on together with the Auto High Beam (AHB) warning indicator.

This warns the driver that the system is activated and ready to automatically switch on the high beam headlamps when the light level falls a little further to approximately 1 lux, as measured by the 'ambient light sensor' located in the mirror body. High beam is generally not required with light levels above 1 lux.

Road Speed

A road speed signal is received by the Body Control Module/Gateway Module (BCM/GWM) from the Anti-lock Braking System (ABS) module via the High Speed (HS) Controller Area Network (CAN) powertrain bus. When the other activation conditions are correct, the BCM/GWM will switch the headlamps to high beam when the road speed has increased above 40 km/h (25 mph).

When the road speed falls to below 24 km/h (15mph), the BCM/GWM will switch the headlamps to low beam. The 10 mph (15 km/h) difference between the on and off road speed thresholds prevents the system continually switching between high and low beam at low speeds.

Override

The driver can manually override the Auto High Beam (AHB) system at any time. When the AHB system is activated, pulling the left steering column multifunction switch to the high beam 'flash' position or pushing it forward to the high beam position will de-activate the system and the AHB warning indicator in the instrument cluster will extinguish.

When the multifunction switch is returned to the central position, from a forward high beam position, the system is re-activated and the high beam assist warning indicator will illuminate again.

Correct Performance

In addition, Auto High Beam (AHB) will only exhibit best performance if all of the following conditions are met:

- No false inputs are received by the camera (image) sensor, such as reflected light from certain static signs – refer to 'System Limitations' section that follows
- Headlamps are correctly aligned
- AHB system has been set for correct 'Hand of Traffic' via the driver menu settings – refer to 'Setting Hand of Traffic' section that follows
- Camera (image) sensor has been through a self learning 'auto aim' calibration procedure if any components have been replaced – refer to 'Calibration' section that follows
- There are no large reflective items, white papers, etc., sitting on top of the dash board in near view of the camera (image) sensor, or stickers placed directly in front of the camera (image) sensor

Driver Menu Features

The Auto High Beam (AHB) feature must first be enabled using the configuration menu available in the instrument cluster. However if required, the AHB system can be permanently disabled leaving the basic 'Autolamps'

system still operative.

Within this menu the system can also be configured for driving on the alternate side of the road (Hand of Traffic). This enables the system to be used in different regions and its' setting is important for correct operation.

Setting 'Hand of Traffic' and High Beam Assist 'Enable'

To set the high beam assist options the following steps must be sequenced:

- With the ignition in power mode 6 (ignition on), and the engine not running, use the controls on the steering wheel to select on the instrument cluster menu:
 - Menu > Vehicle Set-up > High Beam Assist
- Configure the 'Hand of Traffic' setting by selecting the appropriate 'Drive on Left' (of road) or 'Drive on Right' (of road) to the applicable Market condition
- Enable the feature by setting 'Activate Assist' if not already selected.

NOTE:

Enabling or disabling high beam assist will not affect the 'Hand of Traffic' settings once set.

System Limitations

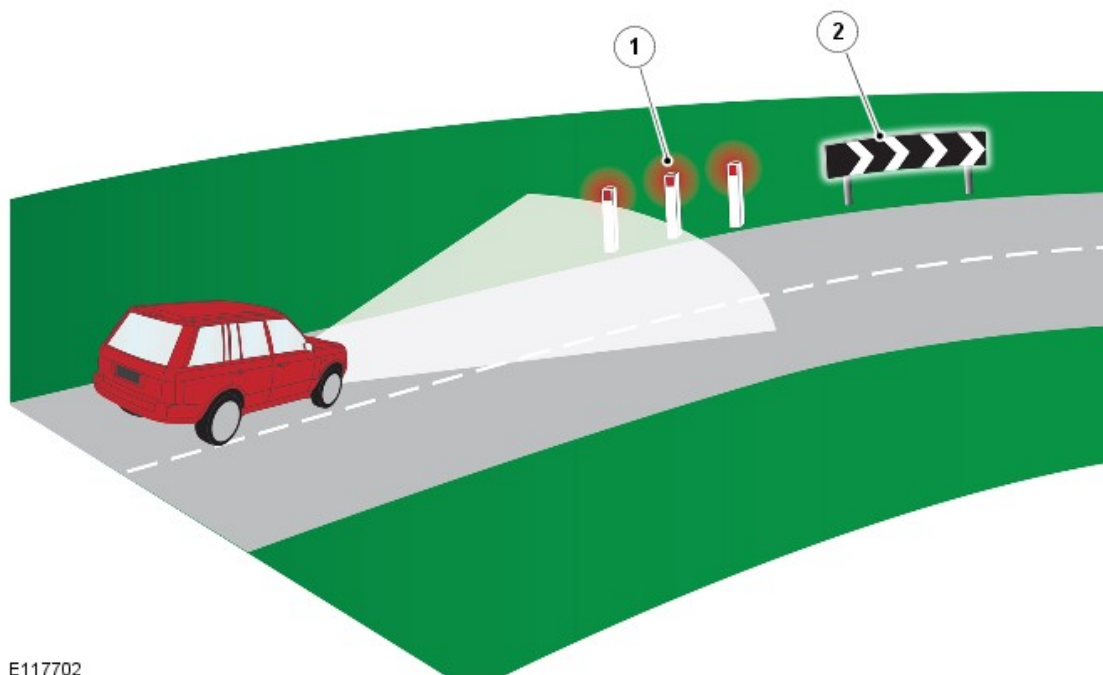
The Auto High Beam (AHB) system can occasionally have difficulty distinguishing between light from other vehicles or reflected light from static highly reflective road signs.

These situations may cause the AHB system to undesirably operate the high beam headlamps or take no action at all. Examples of these situations are as follows:

- Dips, hollows or crests in the road

- Highly reflective static Road signs
- Tight bends
- Poorly illuminated vehicles e.g. cyclists or small mopeds
- Motorway central barriers
- Extreme weather conditions e.g. Fog, heavy snow
- Exterior domestic or industrial lighting

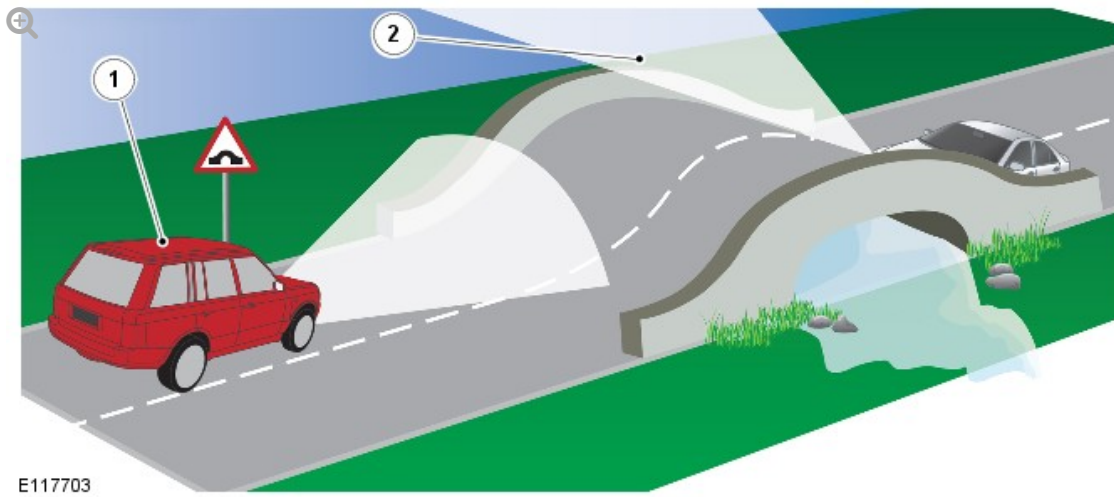
Reflective Static Signs



E117702

ITEM	DESCRIPTION
1	Red reflective signs could be detected as rear tail lamps
2	Large reflective signs could affect the system

Manual Deactivation



ITEM	DESCRIPTION
1	Vehicle equipped with high beam assist
2	Oncoming vehicle headlights can be seen by the driver before the AHB image sensor detects the oncoming light input

There are situations when a driver is able to judge if a high beam deactivation is desirable before the Auto High Beam (AHB) system actually operates, for example over a crest of a hill. Headlamps from an oncoming vehicle can sometimes be seen on the horizon prior to the detection sensor receiving an input. It is the drivers' preference to determine if early intervention is desired in this and similar situations.

System Diagnosis

NOTE:

Windshield stickers, stone chips, dirt and general road film will affect the successful operation of the image sensor if sufficient blocking is present. Avoid placing reflective objects on the instrument panel, for example white paper which can affect the image sensor.

Auto High Beam (AHB) has a self diagnosis capability by comparing data from the ambient light sensor input (located in the rear view mirror) to light

levels detected by the image sensor. If a deviation is detected it is assumed that the ambient light available to the image sensor is being restricted by dirt or other blockage and the system will be deactivated. Diagnostic Trouble Code (DTC)s are stored in the control modules' memory and can be accessed using an approved diagnostic system. Within the diagnostic system is a procedure to test the basic operation of the camera function.

In the event of a fault, the warning strategy to the driver is as follows:

- Image sensor internal fault - green icon will extinguish with no additional message to driver
- Body Control Module/Gateway Module (BCM/GWM) has lost all communication with image sensor - green icon will extinguish with no additional message to driver
- Image sensor blocked - green icon will extinguish with an additional "AHB Sensor Blocked" message within the message centre

System Calibration

To achieve effective operation of the Auto High Beam (AHB) system, a calibration routine is performed on vehicle build and system tolerances are set to an accuracy of +/- 0.2 degrees.

This initial calibration is a 'one time only' procedure. Should the AHB components or the windshield require replacement at the dealership, an automatic calibration routine will be performed. This 'auto aim' calibration procedure is a continual process that takes place during a normal drive cycle at night and could take between 10 - 30 minutes dependant on the following driving conditions:

- If sufficient road markings (lane markings) are visible to the image sensor - approximately 10 minutes
- If insufficient road markings are visible, the system uses the tail lights of preceding vehicles - approximately 30 minutes.

NOTES:

- Until this calibration is complete the system may not react correctly during operation. This should be made clear to the customer before vehicle handover. During any calibration or rectification work the headlamps should be checked for correct alignment.
- Due to mechanical calibration tolerance the correct mirror assembly must be used for the vehicle model types in question and it is not exchangeable with other vehicle model types.
- After any rectification work and before any calibration drives, the headlamps should be checked for correct alignment.

HEADLAMP DELAY

The Body Control Module/Gateway Module (BCM/GWM) controls a headlamp delay function which illuminates the driveway after leaving the vehicle. The headlamp delay will operate on low beam headlamps only regardless of the position of the left hand steering column multifunction switch.

The headlamp delay is activated when the lighting control switch is in the 'Auto' position and the engine is switched off. The message center displays a 'HEADLIGHT DELAY' message and the low beam headlamps will be activated for a period of approximately 30, 60, 120 or 240 seconds, which is selectable using the instrument cluster menu. After the delay period, the BCM/GWM automatically switches off the delay function, extinguishing the headlamps.

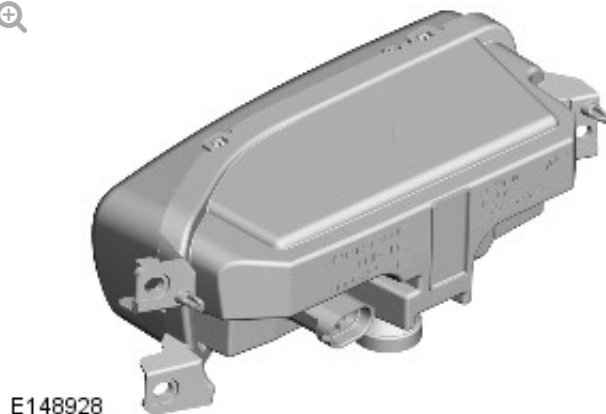
The headlamp delay feature can also be switched off by pressing the headlamp button on the smart key.

INSTRUMENT CLUSTER WARNING INDICATOR ILLUMINATION

The warning indicators in the instrument cluster for left and right turn signal indicator, front and rear fog lamps and headlamp low and high beam and side lamps are activated by the Body Control Module/Gateway Module (BCM/GWM) via messages to the instrument cluster on the Medium Speed (MS) Controller Area Network (CAN) body and High Speed (HS) Controller Area Network (CAN) powertrain buses. The synchronisation of the turn signal warning indicator with the external turn signal indicator lamp frequency is controlled by a cyclic transmission of the light status on the medium speed CAN bus.

The Adaptive Front lighting System (AFS) warning indicator is controlled by a signal from the AFS control module.

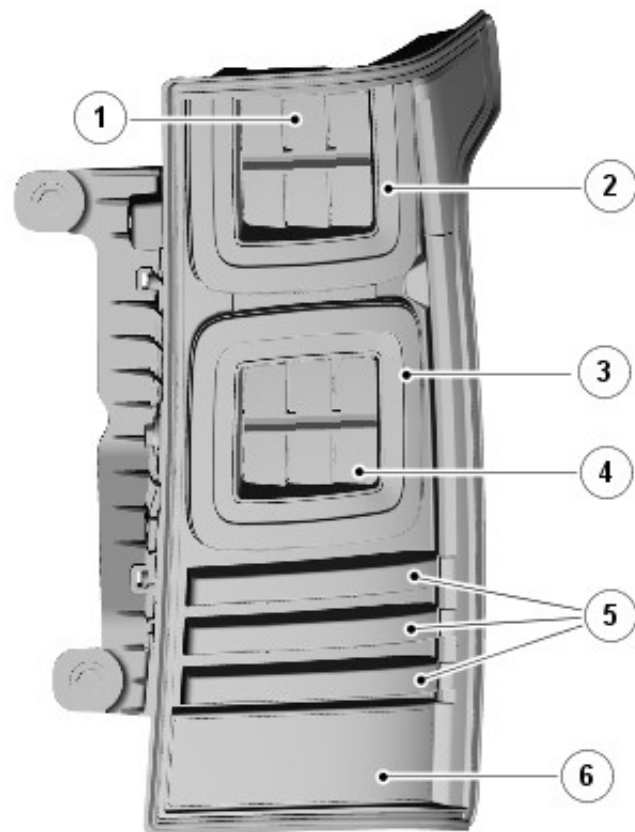
FRONT FOG LAMPS



The front fog lamps are located in the front bumper below the head lamps. Each lamp provides illumination from two Light Emitting Diode (LED)s located in the top of the unit.

Adjustment of the fog lamp units is via an adjuster located in the base of each unit.

TAIL LAMP ASSEMBLY



E148929

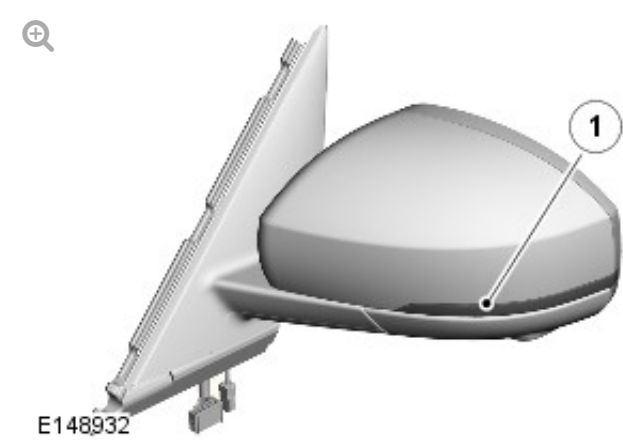
ITEM	DESCRIPTION
1	Stop lamp Light Emitting Diode (LED)s
2	Tail lamp Light Emitting Diode (LED)s
3	Tail lamp Light Emitting Diode (LED)s
4	Reversing lamps
5	Turn signal indicator Light Emitting Diode (LED)s
6	Fog lamps Light Emitting Diode (LED)s

All lamps in the tail lamp assembly use Light Emitting Diode (LED)s.

On North American Specification (NAS) vehicles, an additional LED is used to accommodate the side marker lamp.

None of the LEDs are serviceable items.

SIDE REPEATER LAMPS



ITEM	DESCRIPTION
1	Light Emitting Diode (LED) direction indicator lamp

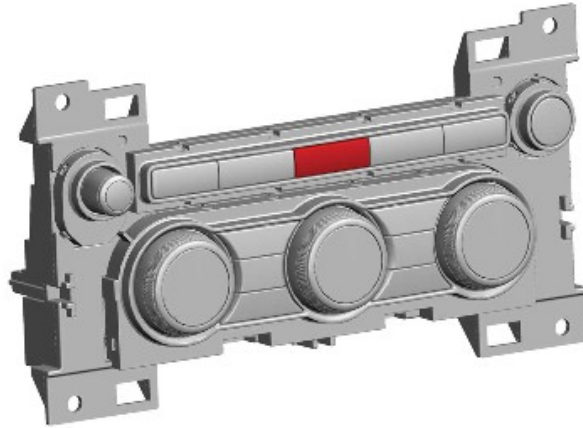
The side repeater lamps are located in the door mirror casing. The side repeater lamps use LEDs for illumination.

LICENSE PLATE LAMPS

Two license plate lamps are located in the trim above the license plate in the lower tail door. The lamps are press fitted in their apertures and secured by an integral plastic clip.

HAZARD WARNING LAMPS

Hazard Warning Switch



E204919

The hazard warning lamps use the front and rear turn signal indicator lamps and the side repeater lamps as previously described. These are controlled by the Body Control Module/Gateway Module (BCM/GWM) in response to a hazard warning lamp request from the instrument panel switch.

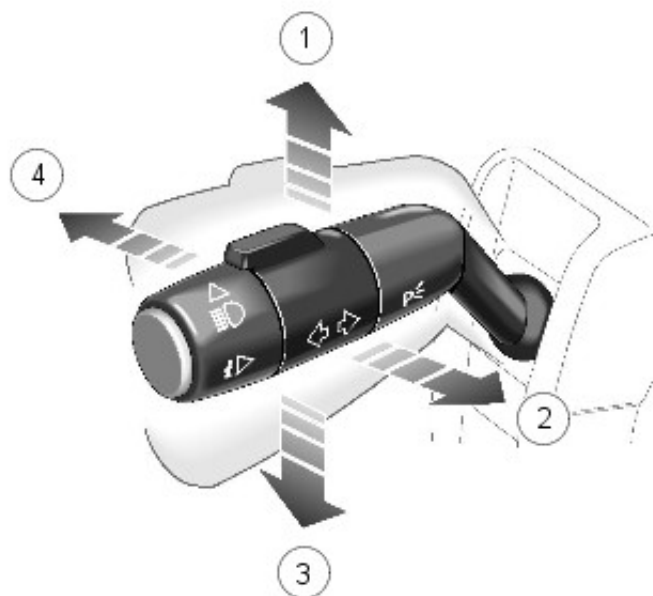
CRASH SIGNAL ACTIVATION

In the event of an accident of a severity to activate and deploy the airbags, the Restraints Control Module (RCM) requests various electrical operations to assist with the crash situation. The RCM requests via the bus systems to the Body Control Module/Gateway Module (BCM/GWM) to activate the hazard warning lamps. These will continue to operate until deselected using the hazard warning lamp switch in the instrument panel. The lamps flash at a frequency which is the same as the frequency used to flash the headlamp high beam in the same situation.

DAYTIME RUNNING LAMPS

Daytime Running Lamps (DRL) operation is detailed in a separate section. For additional information refer to 417-04 Daytime Running Lamps.

STEERING COLUMN MULTIFUNCTION SWITCH

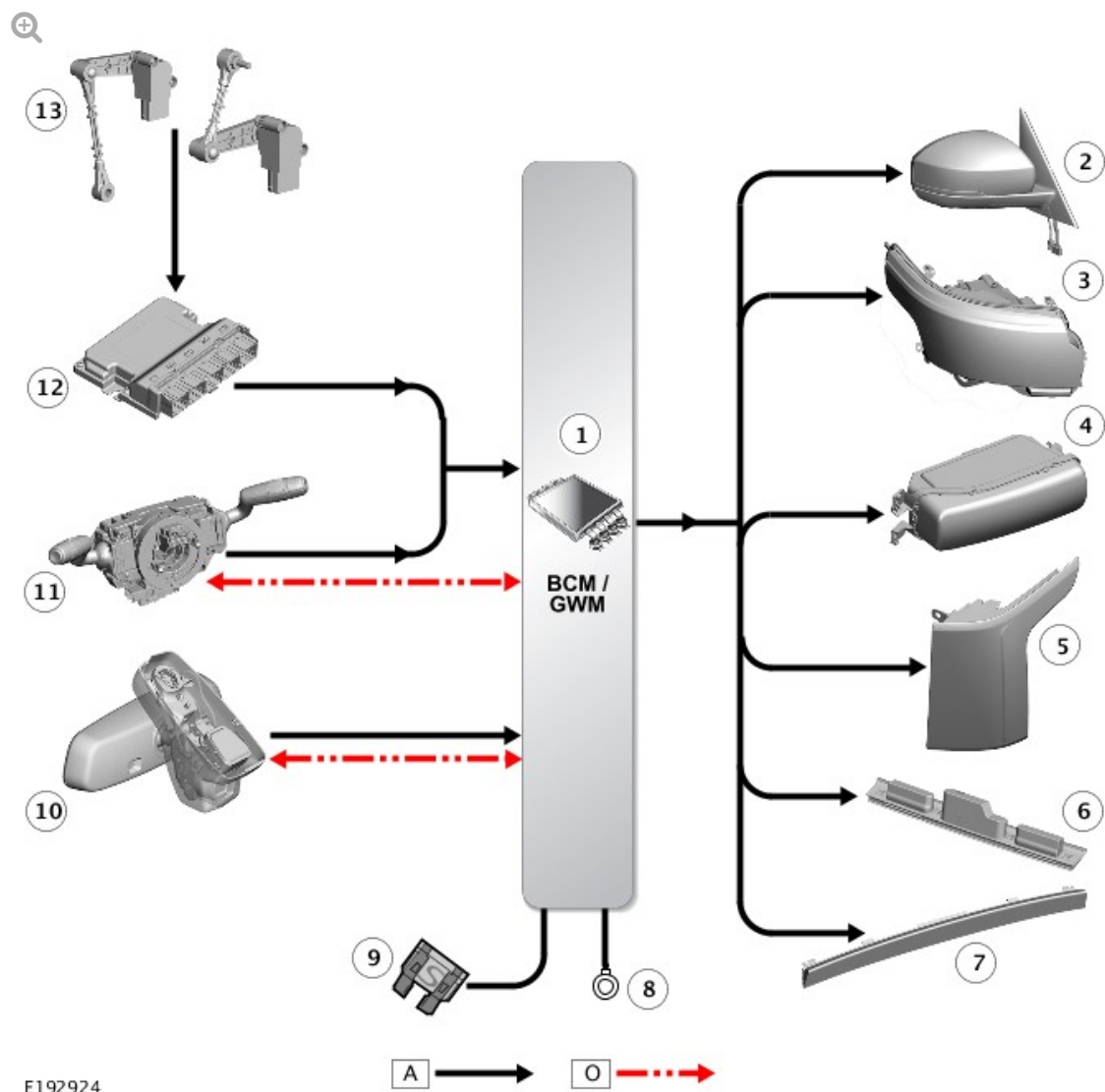


M865859

ITEM	DESCRIPTION
1	Right turn signal indicator
2	Headlamp flash
3	Left turn signal indicator
4	Headlamp high beam

CONTROL DIAGRAM

Exterior Lighting Control Diagram

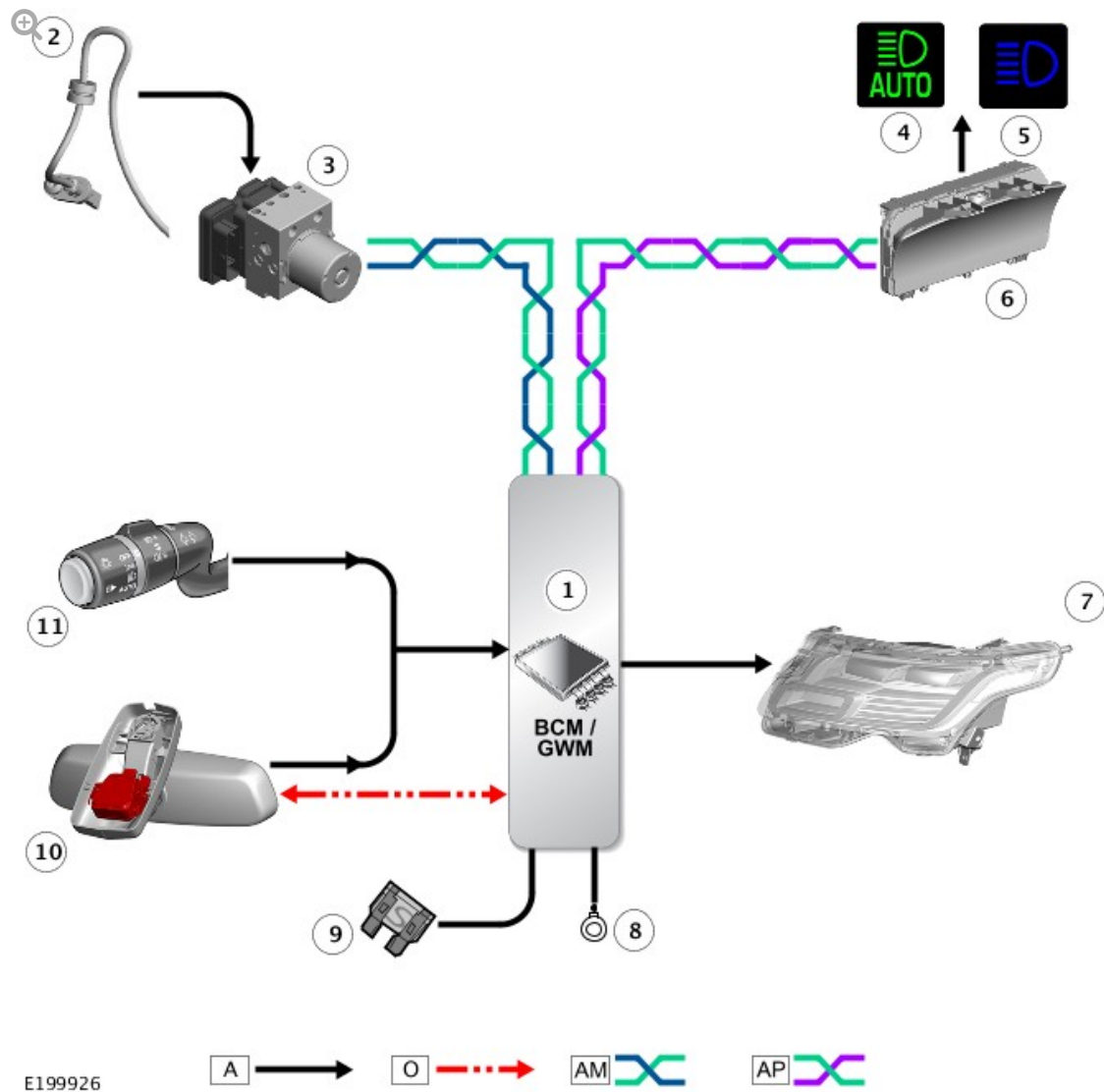


A = HARDWIRED: O = LOCAL INTERCONNECT NETWORK (LIN) BUS.

ITEM	DESCRIPTION
1	Body Control Module/Gateway Module (BCM/GWM)
2	Side repeater lamp in door mirror
3	Headlamp assembly
4	Front fog lamp
5	Tail lamp assembly
6	License plate lamps
7	High Mounted Stop Lamp (HMSL)
8	Ground

9	Fuse
10	Rain/light sensor
11	Clockspring (CLKSPRNG)
12	Chassis Control Module (CHCM)
13	Height sensor

Auto High Beam Control Diagram



A = HARDWIRED: O = LOCAL INTERCONNECT NETWORK (LIN) BUS: AM = HIGH SPEED (HS) CONTROLLER AREA NETWORK (CAN) CHASSIS BUS: AP = MEDIUM SPEED (MS) CAN COMFORT BUS

ITEM	DESCRIPTION

1	Body Control Module/Gateway Module (BCM/GWM)
2	Wheel speed sensor
3	Anti-lock Brake System (ABS) module
4	Auto High Beam (AHB)
5	Headlamp high beam
6	Instrument Cluster (IC)
7	Headlamp assembly
8	Ground
9	Fuse
10	Rain/light sensor, Auto High Beam (AHB) camera
11	Lighting control switch